GCD, LCM, Prime Factorization

1. Warmup Problems

A.1 Find $gcd(630, 1500)$ and $lcm(630, 1500)$.

A.2 What are the gcd and lcm of the numbers $81, 108$, and 135?

A.3 (AMC 12) Let N be the second smallest positive integer that is divisible by every positive integer less than 7. What is the sum of the digits of N?

A.4 (AMC 8) How many positive integer factors of 2020 have more than 3 positive factors? (As an example, 12 has 6 factors, namely $1, 2, 3, 4, 6$, and 12.)

2. Standard Problems

B.1 Suppose that m and n are positive integers such that $200m^2 = 3n^3$. What is the minimum possible value of $m + n$?

B.2 (AMC 12) Let n be the smallest positive integer such that n is divisible by 20, n^2 is a perfect cube, and n^3 is a perfect square. What is the number of digits of n?

B.3 (AMC 10) How many ordered pairs (a, b) of positive integers satisfy the equation

$$a \cdot b + 63 = 20 \cdot \text{lcm}(a, b) + 12 \cdot \text{gcd}(a, b),$$

where $\text{gcd}(a, b)$ denotes the greatest common divisor of a and b, and $\text{lcm}(a, b)$ denotes their least common multiple?

B.4 Let m and n be positive integers satisfying $gcd(m^3, n^2) = 2^2 \cdot 3^2$ and $lcm(m^2, n^3) = 2^4 \cdot 3^4 \cdot 5^3$. Find m and n.

B.5 (AIME) For how many values of k is 12^{12} the least common multiple of the positive integers 6^6, 8^8, and k?

B.6 (AMC 10) For some positive integer n, the number $110n^3$ has 110 positive integer divisors, including 1 and the number $110n^3$. How many positive integer divisors does the number $81n^4$ have?

B.7 (AIME) How many positive divisors of 10^{99} are divisible by 10^{88}?

B.8 (AMC 10) How many positive cubes divide $3! \cdot 5! \cdot 7!$?

B.9 (AMC 10) How many positive integer divisors of 201^9 are perfect squares or perfect cubes (or both)?

B.10 (AMC 12) The number $21! = 51,090,942,171,709,440,000$ has over $60,000$ positive integer divisors. One of them is chosen at random. What is the probability that it is odd?

B.11 (AMC 12) For n a positive integer, let $f(n)$ be the quotient obtained when the sum of all positive divisors of n is divided by n. For example,

$$f(14) = (1+2+7+14) \div 14 = \frac{12}{7}$$

What is $f(768) - f(384)$?

B.12 The only prime factors of n are 2 and 3. If the sum of the positive divisors of n is 1815, find n.

B.13 Let $n = 49000$. Find the sum of the positive divisors of n that are divisible by 4.

B.14 (AMC 12) Let $N = 34 \cdot 34 \cdot 63 \cdot 270$. What is the ratio of the sum of the odd divisors of N to the sum of the even divisors of N?

B.15 What is the greatest integer k such that $80!$ is divisible by 45^k?

B.16 Find the smallest integer n such that there are exactly 40 zeros at the end of the number $n!$.

3. Hard Problems

C.1 (AMC 10) Let $a, b, c,$ and d be positive integers such that $\gcd(a, b) = 24$, $\gcd(b, c) = 36$, $\gcd(c, d) = 54$, and $70 < \gcd(d, a) < 100$. Which of the following must be a divisor of a? (gcd means greatest common factor)

(A) 5 **(B)** 7 **(C)** 11 **(D)** 13 **(E)** 17

C.2 (AMC 10) Let n be the least positive integer greater than 1000 for which

$$\gcd(63, n+120) = 21 \quad \text{and} \quad \gcd(n+63, 120) = 60.$$

What is the sum of the digits of n?

C.3 (AMC 12) Suppose a, b, c are positive integers such that

$$a + b + c = 23$$

and

$$\gcd(a, b) + \gcd(b, c) + \gcd(c, a) = 9.$$

What is the sum of all possible distinct values of $a^2 + b^2 + c^2$?

C.4 (AMC 12) Suppose that a, b, c and d are positive integers satisfying all of the following relations.

$$\begin{aligned}
abcd &= 2^6 \cdot 3^9 \cdot 5^7 \\
\operatorname{lcm}(a, b) &= 2^3 \cdot 3^2 \cdot 5^3 \\
\operatorname{lcm}(a, c) &= 2^3 \cdot 3^3 \cdot 5^3 \\
\operatorname{lcm}(a, d) &= 2^3 \cdot 3^3 \cdot 5^3 \\
\operatorname{lcm}(b, c) &= 2^1 \cdot 3^3 \cdot 5^2 \\
\operatorname{lcm}(b, d) &= 2^2 \cdot 3^3 \cdot 5^2 \\
\operatorname{lcm}(c, d) &= 2^2 \cdot 3^3 \cdot 5^2
\end{aligned}$$

What is $\gcd(a, b, c, d)$?

C.5 (AMC 12) How many positive integers n are there such that n is a multiple of 5, and the least common multiple of 5! and n equals 5 times the greatest common divisor of 10! and n?

C.6 (AIME) Let $[r, s]$ denote the least common multiple of positive integers r and s. Find the number of ordered triples (a, b, c) of positive integers for which $[a, b] = 1000$, $[b, c] = 2000$, and $[c, a] = 2000$.

C.7 (AIME) What is the largest positive integer n for which $n^3 + 100$ is divisible by $n + 10$?

C.8 How many integers from 1 to 100 (inclusive) have odd number of positive divisors?

4. Harder

D.1 (AIME) The numbers in the sequence 101, 104, 109, 116,... are of the form $a_n = 100 + n^2$, where $n = 1, 2, 3, \ldots$ For each n, let d_n be the greatest common divisor of a_n and a_{n+1}. Find the maximum value of d_n as n ranges through the positive integers.

D.2 (IMO 1959) Prove that the fraction $\frac{21n+4}{14n+3}$ is irreducible for every natural number n, i.e. $gcd(21n + 4, 14n + 3) = 1$.

D.3 Find the gcd of $2^{19} + 1$ and $2^{96} + 1$.

D.4 Find all possible values of $gcd(2n - 1, 9n + 4)$.

Theorem (Euclidean Algorithm)

To find the *gcd* of two positive integers $a \geq b$, we can repeatedly use the fact that

$$a = qb + r \Longrightarrow gcd(a, b) = gcd(b, r).$$

Note that with this procedure we can replace the pair of numbers (a, b) with a smaller pair (b, r) at every step until *gcd* becomes trivial to compute.

Example 1

Let's calculate $gcd(42823, 6409)$ using Euclidean algorithm.

$$\begin{aligned} 42823 &= 6 \times 6409 + 4369 & gcd(42823, 6409) &= gcd(6409, 4369) \\ 6409 &= 1 \times 4369 + 2040 & gcd(6409, 4369) &= gcd(4369, 2040) \\ 4369 &= 2 \times 2040 + 289 & gcd(4369, 2040) &= gcd(2040, 289) \\ 2040 &= 7 \times 289 + 17 & gcd(2040, 289) &= gcd(289, 17) \\ 289 &= 17 \times 17 + 0 & gcd(289, 17) &= gcd(17, 0) = 17 \end{aligned}$$

Theorem (Fundamental Theorem of Arithmetic)

Every integer $n \geq 2$ has a prime factorization

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

where $p_1, p_2, \dots, p_k$ are distinct primes and $\alpha_1, \alpha_2, \dots, \alpha_k$ are positive integers.

Fact

Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ and $b = p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}$ be the prime factorizations of two positive integers. (Note that two integers don't have to have the same primes in their prime factorizations, such as $a = 2^2 \cdot 5^3$ and $b = 3^3 \cdot 5^1$; but we can can overcome this by writing $a = 2^2 \cdot 3^0 \cdot 5^1$ and $b = 2^0 \cdot 3^3 \cdot 5^1$)

- $a \mid b \iff \alpha_1 \leq \beta_2, \alpha_2 \leq \beta_2, \dots, \alpha_k \leq \beta_k$
- $gcd(a, b) = p_1^{min(\alpha_1, \beta_1)} p_2^{min(\alpha_2, \beta_2)} \dots p_k^{min(\alpha_k, \beta_k)}$
- $lcm(a, b) = p_1^{max(\alpha_1, \beta_1)} p_2^{max(\alpha_2, \beta_2)} \dots p_k^{max(\alpha_k, \beta_k)}$

Theorem (Number of divisors)

The number of the positive divisors of the integer $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ is

$$(1 + \alpha_1) \cdot (1 + \alpha_2) \cdot \ldots \cdot (1 + \alpha_k)$$

Theorem (Sum of divisors)

The sum of all positive divisors of the integer $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ is

$$(1 + p_1 + p_1^2 + \ldots + p_1^{\alpha_1}) \cdot (1 + p_2 + p_2^2 + \ldots + p_2^{\alpha_2}) \cdot \ldots \cdot (1 + p_k + p_k^2 + \ldots + p_k^{\alpha_k})$$

which is equal to

$$\frac{p_1^{\alpha_1+1}}{p_1 - 1} \cdot \frac{p_2^{\alpha_2+1}}{p_2 - 1} \cdot \ldots \cdot \frac{p_k^{\alpha_k+1}}{p_k - 1}.$$